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Review

Food Waste Anaerobic Digestion Under High Organic Loading Rate: Inhibiting Factors, Mechanisms, and Mitigation Strategies

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Abstract

Anaerobic digestion (AD) for food waste (FW) treatment has faced many challenges, especially ammonia nitrogen, acid, and salinity inhibition at a high organic loading rate (OLR). Therefore, a systematic understanding of the issues arising during the FW AD process is a necessity under a high OLR (over 3 g-VS/L d). Primarily, in terms of ammonia nitrogen inhibition, ammonia ions inhibit methane synthesis enzymes, and free ammonia (FAN) contributes to the imbalance of microbial protons. Regulation strategies include substrate C/N ratio regulation, microbial domestication, and ammonia nitrogen removal. In addition, with regard to acid inhibition, including volatile fatty acid (VFA) and long-chain fatty acid (LCFA) accumulation, the elevated acid concentration can contribute to reactive oxygen species stress, and a solution to this includes the addition of alkaline agents and trace elements or the use of microbial electrochemical and biofortification technology and micro-aeration-based AD technology. Furthermore, in terms of salinity inhibition, high salinity can result in a rapid increase in cell osmotic pressure, which can cause cell rupture, and water washing and bio-electrochemical AD are defined as solutions. Future research directions are proposed, mainly in terms of avoiding the introduction of novel containments into these regulation strategies and applying them in large-scale AD plants under a high OLR.

Keywords: anaerobic digestion; high organic loading rate; food waste; ammonia inhibition; acid inhibition; salt inhibition

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1. Introduction

FW has been recognized as a widespread issue that transcends socioeconomic boundaries [1]. Specifically, the Food and Agriculture Organization of the United Nations has defined FW as any alteration in the availability, palatability, hygiene, or quality of edible materials that hinders their consumption. In addition, according to data from the National Bureau of Statistics of China, the generation of FW has increased from the year 2012 to 2023, with a peak in the year 2022. Such a huge amount of food waste is bound to cause many environmental problems and resource waste. Therefore, both solving environmental problems and resource recycling should be controlled during food waste treatment. Therefore, FW is defined as a complex issue due to its intricate composition, high moisture,

and organic content [2]. Leachate from landfills requires additional treatment, increasing sewage treatment costs; fly ash generated by incineration is classified as hazardous waste, and improper disposal can cause secondary pollution. Treatment methods only regard food waste as "waste", failing to achieve the conversion and utilization of organic matter, with an extremely low resource recovery efficiency (for example, incineration only generates thermal energy, and the utilization rate is less than 30%). Such approaches are generally not suitable for systematic application in FW treatment [3,4]. Therefore, these two approaches are not in line with current development trends [5]. Conversely, technologies such as AD and composting, which can convert FW to high-value-added products, have been recognized as methods for efficient FW resource utilization [5,6]. Among the above-mentioned approaches, the AD process has been assumed to be a relatively mature technology for decomposing organic waste because anaerobic microorganisms can convert organic compounds into biogas [7]. Specifically, the produced biogas contains methane, ranging from 65% to 75%, and carbon dioxide, ranging from 25% to 35% [8]. Thus, AD presents the advantages of minimal carbon emissions, reduced secondary pollution and economical operational expenses. Furthermore, it is suitable for centralized, large-scale, organic waste treatment [9]. As an alternative environmental technology, AD has been increasingly utilized for various types of organic waste treatment, particularly FW. Nevertheless, despite it having numerous advantages, the widespread industrialization of the AD process can be hindered by inferior operational stability and a low OLR [10].

In general, as a biological process for organic compound degradation, the AD process can be divided into four primary steps, including hydrolysis, acidogenesis, acetogenesis, and methanogenesis in the absence of free molecular oxygen [7]. According to previous studies, the hydrolysis step can be generally recognized as the rate-limited step for complicated organic substrate biodegradation, caused by the formation of toxic by-products (complex heterocyclic compounds) or the accumulation of long-chain fatty acids (LCFAs) and volatile fatty acids (VFAs) [11,12]. Therefore, accompanied with an increase in the OLR, acid accumulation can be aggravated, which can contribute to bioreactor deterioration. Furthermore, although ammonia can play an important role in balancing the C/N ratio in an AD system, the excessive addition of ammonia can contribute to FAN accumulation, which can impede FW decomposition and methane production cessation [13]. In addition, salinity has been considered indispensable during food production. Nevertheless, high salinity can influence the diversity of the sludge, thereby suppressing the activity of methane-producing archaea (MPA), and eventually lead to system collapse [14]. Therefore, in terms of the AD process for FW treatment under a high OLR, the intermediate products and by-products produced (FAN and VFAs), as well as the high salinity caused by the high OLR of the substrate, are all essential to the performance of the AD process [15]. In actual AD plants for FW treatment, the OLR was maintained at around 2–3 g-VS/L d as usual [16], while for previous lab-scale FW AD bioreactors, the OLR could be increased to over 10 g-VS/L d because most of these challenges could be mitigated by the introduction of a novel bioreactor [17,18]. However, due to the different inhibition mechanisms of ammonia nitrogen, acid, and salinity under a high OLR, solutions to alleviate these inhibitions are different. Therefore, the mechanisms of the above-mentioned three aspects and regulation strategies should be aimed at alleviating ammonia nitrogen, acid, and salinity inhibition.

The purpose of this review is to provide a comprehensive overview of these three inhibition factors during the FW AD process under a high OLR and their regulation strategies. Initially, the mechanisms of ammonia nitrogen inhibition were analyzed, and regulation strategies were proposed. Subsequently, mechanisms of acid and salinity inhibition were analyzed, respectively, and the solutions were proposed, respectively. Furthermore, research gaps and opportunities were provided to investigate the advantages and disadvantages

of each strategy. Additionally, research gaps and opportunities are identified to critically evaluate the merits and limitations of each strategy. Lastly, future research directions are proposed, focusing primarily on the large-scale practical implementation of these regulatory strategies in AD treatment plants to bridge the gap between laboratory findings and industrial application.

2. Organic Loading Rate

The OLR (g-VS/L d) can be defined as the quantity of influent organic compounds per liter of AD reactor, and it is one of the most crucial indicators to ensure the efficient operation of the AD process [2,19]. AD bioreactors operating under a high OLR can decrease the capacity requirement, as well as enrich functional bacterial species [20]. Nevertheless, accompanied by the increase in OLR, there is a risk of VFA and LCFA accumulation, as well as ammonia and high-salinity inhibition due to the combined effects of acid, ammonia, salinity, unequal distribution during mixing, and poor heat transfer. These conditions have the potential to cause irreversible failures in AD bioreactors, including gas production and sludge acidification [21]. As illustrated in Table 1, continuous-stirred tank reactors (CSTRs) are capable of maintaining at a high OLR of approximately 5–6 g-VS/L d on average, which is relatively lower compared to other types of anaerobic reactors. Conversely, in the case of other reactors, such as two-stage reactors, three-stage reactors, and siphon-driven self-agitated reactors (SDSARs), a high OLR is defined as above 7 g-VS/L d. In practical cases, to ensure the stable operation of plants, the OLR is typically maintained at a low level (<3 g-VS/L d). Therefore, the term "high OLR" for an FW AD bioreactor can be defined as a situation in which the OLR exceeds 3 g-VS/L d according to previous studies [22].

Table 1. Typical high OLR values for different AD bioreactors for FW treatment.

Process	Reactor Volume and Type	Operation Temperature	Operation Condition	High OLR (g VS/L d)	Ref.
Single-stage	2 L semi-continuous CSTR	Mesophilic	OLR: 0.5–6, HRT: 200–6.7	3	[23]
Two-stage	0.5 L acidogenic reactor and 2 L semi-continuous CSTR	Mesophilic	OLR: 0.5–5, HRT: 200–20	4	
Single-stage	230 L CSTR	Thermophilic	OLR: 3.57 HRT: 20	6.0–7.0	[24]
Two-stage	200 L and 760 L CSTR	Thermophilic	O OLR: 3.5–7 HRT: 20	6.0–7.0	
Three-stage	1 L and 20 L semi-continuous reactor	Mesophilic	OLR: 1.6–10 pH: 5.1–7.8	10	[17]
Single-stage	5000 L reactor	Mesophilic	OLR: 0–37	6.4	[25]
Two-stage	5000 L and 5 L CSTR reactors		OLR: 0–20	8.3	
Single-stage	6 L CSTR	Mesophilic	OLR: 2.4, HRT: 30 pH: 7.77	4	[26]
Temperature-phase two-stage	1.5 L thermophilic CSTR 6 L mesophilic CSTR		OLR: 14.2 and 2.6 HRT: 3 and 12 pH: 5.36 and 7.59	6.3	
Single-stage	60 L typical single-stage vertical CSTR reactor	Mesophilic	OLR: 0.545–10.3 pH: 7	5.8	[27]
Two-stage	27 L novel semi-continuous reactor		OLR: 0.545–10.3 HRT: 3 and 12 pH: 5.36 and 7.59	6.9	
Single-stage	3 L thermophilic	Mesophilic	OLR: 4.2–10	7.3	[28]
Single-stage	10 L SDSAR	Mesophilic	OLR: 3.0, 4.8, 7.3, 7.3, 14.4	-	[18]
	10 L SDSAR	Thermophilic	OLR: 3.0, 4.8, 7.3, 7.3	10	

3. Ammonia Inhibition: Mechanisms and Mitigation Strategies

3.1. Ammonia Inhibition

Typical FW can be characterized by a low C/N ratio, high protein, and nitrogen content. Therefore, nitrogenous organic compounds such as proteins and amino and nucleic acids can be hydrolyzed into ammonia nitrogen, which can easily accumulate in AD bioreactors. Based on the aforementioned challenge, it is crucial to control its concentration because surplus ammonia nitrogen is a potential hindrance on the decomposition of organic substrates and may reduce methane production or even destabilize the system [29,30]. Conversely, ammonia nitrogen plays a crucial role as a nutrient for the vital functions of anaerobic microorganisms. Specifically, ammonia nitrogen contributes to partial alkalinity and the buffering capacity, which can counteract a low buffering capacity resulting from excessive acid accumulation. In addition, previous studies have demonstrated that high levels of ammonia nitrogen seriously inhibit the activities of methane-producing archaea (MPA), leading to an imbalance in the AD system due to synergistic interactions among multiple microorganisms [29,31]. In terms of the microbial community, MPA were the predominant microorganisms that were inhibited by elevated ammonia levels. This phenomenon was particularly evident for acetate-utilizing MPA. In contrast, hydrolytic bacteria demonstrated high resistance to the adverse effects of high ammonia stress [32–34]. For instance, Wang et al. investigated the response of hydrogen-utilizing and acetate-utilizing MPA to different ammonia nitrogen concentrations under mesophilic conditions and revealed that hydrogen-utilizing MPA were 11 and 3 times more tolerant than acetate-utilizing MPA, respectively [34]. In addition, Chen et al. found that when the concentration of ammonia nitrogen in the AD reactor exceeded 2000 mg/L, methane production decreased significantly. Microbial community analysis revealed that a high concentration of ammonia nitrogen mainly caused a significant decrease in the abundance of specialized acetate-utilizing MPA such as the Methanosaeta, while there was no significant effect on hydrogen-utilizing MPA [35]. Similar conclusions were obtained from the studies by Christou et al. and He et al. as well [36,37].

Nevertheless, FAN has been identified as the primary factor contributing to the inhibition of the AD system due to its permeability [38]. Previous studies have demonstrated that ammonia inhibition can impact MPA through two primary aspects: (1) ammonia ions can directly inhibit the activity of enzymes essential for methane synthesis. According to previous research, Kadam and Boone found that the tolerance of three enzymes involved in methanogenesis, glutamate dehydrogenase, glutamine synthetase, and alanine dehydrogenase, varied in response to ammonia nitrogen levels [39]. (2) Hydrophobic FAN can lead to proton imbalance or intracellular K^+ deficiency, as shown in Figure 1. Hydrophobic FAN can penetrate microbial cells through passive diffusion, increasing intracellular osmotic pressure. To maintain osmotic balance between the inside and outside of the cell, it actively transports K^+ out of the cell, eventually leading to proton imbalance or inactivation due to intracellular K^+ deficiency. Furthermore, Sprott conducted a study involving bacteria cultured in an ammonia solution without K^+, which revealed a loss of approximately 98% of K^+ through exchange reactions. In addition, FAN contributed to osmotic pressure imbalance across the cell membrane, further contributing to decreased microbial activity [40].

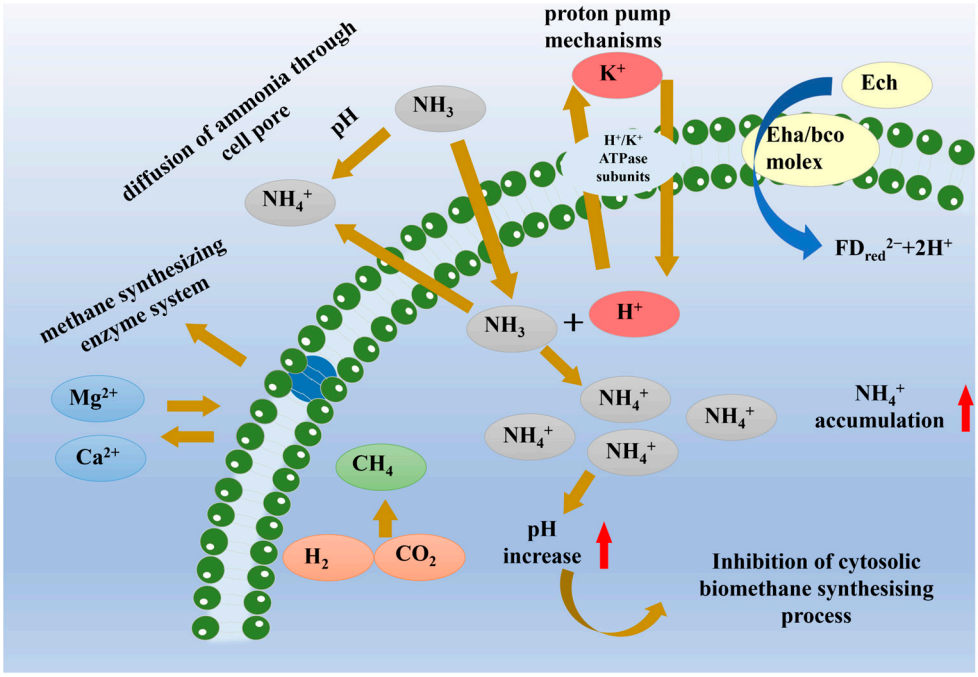


Figure 1. Mechanism of cell activity inhibition by ammonia nitrogen.

In terms of ammonia nitrogen concentration, it has been documented in previous research that lower concentrations (50–200 mg/L) are typically conducive to the AD system because ammonia nitrogen can serve as an essential source of nutrients for anaerobic microorganisms involved in nucleic acid, amino acid, and protein synthesis [29]. Additionally, ammonia nitrogen has a certain buffering capacity that is beneficial for maintaining neutral pH conditions [41]. For instance, mesophilic AD bioreactors are most effective for FW treatment when the total ammonia nitrogen concentration ranges from 600 to 800 mg/L and the pH level ranges from 7.2 to 7.5 [29]. Conversely, high ammonia concentrations (>1500 mg/L) have been identified as primary factors contributing to destabilization in AD systems because they can cause a severe inhibition of organic substrate degradation and significant decreases in biogas production [29]. Furthermore, according to a previous study, over 2000 mg/L ammonia nitrogen in a semi-continuous FW AD reactor cannot only lead to VFA accumulation and a pH decrease but also suppress MPA activity and the acetoclastic methanogenesis pathway. Syntrophic acetate oxidation coupled with hydrogenotrophic methanogenesis pathways can be enhanced as well [42]. In addition, another study demonstrated that bacteria involved in acetate production exhibited heightened sensitivity to ammonia nitrogen stress when the FAN concentration exceeded 200 mg/L in an upflow anaerobic sludge blanket (UASB) [43]. Moreover, when the total ammonia nitrogen concentration exceeded 4300 mg/L, the activity of cellulose-hydrolyzing bacteria and MPA decreased because dominant acetoclastic MPA such as Methanosaeta in seed sludge noticeably reduced their activities during cultivation. This decline can be attributed to various factors including the substrate type, inoculum quality, temperature changes, and pH transformation [43]. The concentration of ammonia inhibition in AD with different substrates is summarized in Table 2.

Table 2. Concentrations of ammonia inhibition in FW AD bioreactor.

Organic Loading Rate	Temperature	Concentration (mg/L)	Inhibition Principle	Ref.
1.8 g-VS/L d	35 ± 1 °C	TAN: 2400	The expression levels of acetogenesis, butyrate degradation, propionate degradation, and methane production were significantly inhibited by a high NH_4^+ concentration	[44]
5.87 g-VS/L d	37 ± 1 °C	TAN: 6500	High TAN levels showed toxicity on Methanosarcina in TAN systems	[45]
2.5 g-COD/L d	37.5 ± 0.17 °C 22.6 ± 0.82 °C	TAN: 1800 TAN: 3000	The hydrogenotrophic MPA were dominant when the TAN concentrations were high	[46]
2.72 g-COD/L d	35 ± 1 °C	TAN: 1750	The expression levels of propionate degradation and methane production were significantly inhibited by a high NH_4^+ concentration	[47]
0.8 g-COD/L d	35 ± 1 °C	FAN: 290	The expression levels of methanogenesis were significantly inhibited by a high FAN concentration	[43]
0.15 g-COD/L d	35 ± 1 °C	TAN: 16,000	The hydrogenotrophic MPA were dominant when the TAN	[48]
1.0 g-VS/L d	37 ± 1 °C	TAN: 2100	inhibition of ammonia on syntrophic acetogenesis was caused by a population decrease in both active syntrophics	[49]
2 g-COD/L d	37 ± 1 °C	FAN: 500	The activity of acetoclastic MPA was completely inhibited	[34]

Nevertheless, the inhibition of methanogenesis cannot be recognized as a corollary due to ammonia nitrogen accumulation. During the AD process, hydrolysis, acidification, and hydrogenotrophic methanogenesis are comparatively less susceptible to a high ammonia nitrogen concentration because these processes require a high bacterial diversity and abundance. Conversely, acetoclastic MPA are sensitive to high ammonia nitrogen concentrations because these can contribute to gas production cessation and acetate accumulation during the start-up phase of the AD system [50,51]. Nevertheless, dominant species of acetoclastic MPA can transform into ammonia nitrogen-tolerant genera and mutualistic acetic acid-oxidizing bacteria, which can facilitate the restoration of acetate metabolism and normal methanogenesis procedures [42]. Meanwhile, when ammonia-tolerant sludge is inhibited by high concentrations of ammonia nitrogen, acetate oxidation occurs, and the acetoclastic methanogenesis pathway can be converted into a hydrogenotrophic pathway. This serves as a response mechanism that ensures the conversion of acetate to methane [48]. Therefore, it can be concluded that acetoclastic methanogenesis is not the primary factor responsible for destabilization caused by ammonia inhibition. Due to variability in experimental conditions and differences in substrate composition, further research is needed to identify the precise limiting steps and detailed reasons for the inhibition of FW AD systems by ammonia nitrogen.

3.2. Regulation Strategies of AD System Under High Ammonia Nitrogen

In general, high concentrations of ammonia nitrogen have been considered a typical inhibitor in FW AD systems. Therefore, regulation strategies should be adapted to monitor appropriate ammonia nitrogen concentrations. Typical regulation strategies include substrate C/N ratio regulation, microbial domestication, and ammonia nitrogen removal.

3.2.1. Substrate C/N Ratio Regulation

In general, maintaining the inoculated C/N ratio between 20 and 30 has been considered an appropriate condition for biological treatments such as AD systems [52]. The high

C/N ratio of substrate can contribute to an insufficient nitrogen source, which can impede complete utilization of the carbon source. Conversely, a low C/N ratio can contribute to ammonia accumulation and inhibition of anaerobic microorganisms' activity [53]. The C/N ratio of FW is generally greater than 30, and substances with a relatively low C/N can be co-digested to achieve the optimal C/N ratio. For example, in a previous study by Zhai et al., the optimal mixing ratio for co-digestion of FW (C/N ratio 31.18) and cow manure (C/N ratio 22.37) was determined to be 1:1. This ratio resulted in a maximum methane yield of 179.8 mL/g-VS at an initial pH of 7.5 [54]. In addition, the co-digestion of FW and cattle manure (mixing ratios of 0.47:1, 0.32:1, and 0.16:1, respectively) resulted in higher specific methane yields than mono-digestion at 5 g-VS/L, 4 g-VS/L, and 3 g-VS/L of OLR. The co-digestion of FW and cattle manure at their respective mixing ratios resulted in the following methane yields: 358, 444.7 and 402 mL/g-VS [55]. Therefore, the co-digestion of substrates with a high C/N ratio and FW is defined as the most effective approach for the efficient operation of AD systems. According to previous research, Peng et al. observed that a CSTR cultivated with unitary FW exhibited significant ammonia suppression, which brought about a 92.80% reduction in methane yield. Nevertheless, ammonia suppression was alleviated when 40% of the sludge was substituted with landfill leachate due to an enrichment of bacteria, such as the genus Pelotomaculum, which has a higher relative abundance [56]. However, the mixing ratio needs to be accurately adjusted; otherwise, it may aggravate ammonia accumulation or nutritional imbalance. In addition, introducing exogenous substrates may introduce heavy metals (such as Pb^{2+}), which are difficult for the system to evaluate.

3.2.2. Microbial Domestication

In general, high concentration of ammonia nitrogen directly affect the activity of anaerobic microorganisms, particularly MPA. Zhang et al. identified that the tolerance ranges of anaerobic microorganisms were from 1700 to 7000 mg/L for total nitrogen and from 53 to 1450 mg/L for FAN, respectively [57]. According to the characteristics of microorganisms' adaptability to external environments, MPA can be acclimated by increasing the ammonia nitrogen concentration in an anaerobic bioreactor step by step, which in turn improves their tolerance to ammonia nitrogen. On the other hand, domesticated microorganisms can be introduced into the reactor as functional microbial flocs, which can enhance their ammonia nitrogen tolerance capacity and further improve the stability of AD systems. For example, Fotidis et al. found that the domestication of anaerobic microorganisms through the gradual addition of ammonia concentrations ranging from 1 to 7 g can increase the tolerance limit of the ammonia concentration from 5000 to 7000 mg/L [58]. In addition, Gao et al. utilized FW as a substrate to increase the total ammonia tolerance limit for acclimatizing anaerobic microorganisms up to 4293 mg/L [59]. It is worth noting that the domestication period generally lasted for several weeks (>30 days), during which fluctuations in reactor efficiency led to gas production loss. In addition, merely improving the tolerance of specific bacterial communities was insufficient in the face of complex inhibitory factors (such as the synergistic effect of ammonia nitrogen and salinity). Furthermore, recent research has indicated that the microbiome and enzymes can be modulated by the addition of glutathione. This process significantly up-regulated five key proteins related to the ribosome and promoted microbial proliferation and enzyme synthesis through ammonia, which contributed to improving the total ammonia nitrogen tolerance limit. Additionally, key proteins involved in aminoacyl-tRNA biosynthesis (i.e., aspartyl-tRNA synthetase, methionyl-tRNA synthetase, glutaminyl- tRNA synthetase, and lysyl-tRNA synthetase) were significantly up-regulated by glutathione supplementation. It was hypothesized that

glutathione-induced up-regulation of key proteins in transcription and translation could result in more efficient enzyme synthesis (Figure 2) [60].

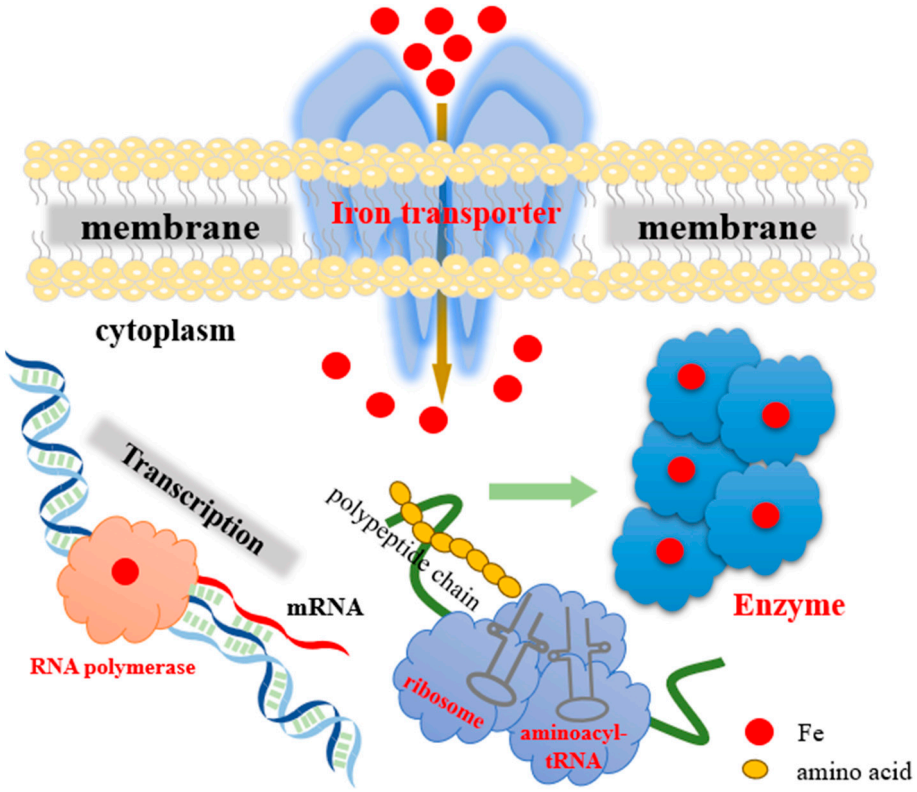


Figure 2. The schematic diagram of microbial enzyme synthesis being enhanced by glutathione under ammonia stress (the bold red fonts represent the parts being improved by glutathione).

3.2.3. Ammonia Nitrogen Removal

The mitigation of generated ammonia nitrogen through appropriate methods, including the addition of additives and ammonia blowing off, is another effective way to alleviate ammonia inhibition in AD systems, which can significantly reduce the concentration of ammonia nitrogen in a short period. In terms of additives, they can not only exhibit a strong ion exchange ability but also improve the attachment state of microorganisms and substrates. For instance, the addition of nanoscale zero-valent iron sulfide (S-nZVI) observably promoted the enhancement of nanoscale zero-valent iron (nZVI) in consolidating hydrogenotrophic methanogenesis. Furthermore, microbial composition analysis revealed that S-nZVI addition enriched species related to biohydrogen production such as the genus Prevotella and hydrogen-utilizing MPA, including Methanoculleus, thereby accelerating methane production [61]. Another study indicated that incorporating FW anaerobic digestate-derived biochar into the reactor could effectively adsorb NH_4^+ to alleviate ammonia nitrogen accumulation and mitigate its toxicity. Aligned with the decrease in ammonia stress, the structure of the microbial community was further optimized, and the abundance of acetate-producing bacteria involved in butyrate and propionate metabolism was enriched [62]. Nevertheless, excessive addition of additives places requirements on the clogging resistance and piping system of the reactor. Shi et al. conducted experiments with two 6 L conventional anaerobic reactors for FW AD treatment, and they found that the reactor equipped with a gas membrane absorption system could be operated at a higher OLR to eliminate the inhibition caused by ammonia nitrogen [63]. Ammonia blowing off is another typical method to remove ammonia nitrogen, where FAN can be removed by simultaneously increasing the pH and temperature combined with constant stirring.

Among these methods, side-stream blowdown is an effective way to remove ammonia nitrogen [64].

4. Acid

4.1. Acid Suppression

4.1.1. VFA Inhibition

The AD process involves a series of biochemical reactions that convert complex organic compounds into biogas, including methane and carbon dioxide. Acidification is a critical step because complex organic compounds can be converted into small molecular soluble chemical oxygen demand (COD), which further generates VFAs in the acetogenesis phase. Subsequently, the generated VFAs can be converted into biogas during the methanogenesis phase. VFAs mainly consist of formic, acetic, propionic, butyric, and valeric acid and their isomers, which have fewer than six carbon atoms in their carbon chains. Inhibition of methanogenesis due to VFA accumulation often occurs under a high OLR for rapidly hydrolysable organic compounds. VFA accumulation triggers reactive oxygen species (ROS) stress (e.g., superoxide anions), which induces lipid peroxidation and membrane damage (Figure 3), directly inhibiting methanogenesis under a high OLR [65]. According to previous research, Sun et al. demonstrated that MPA activity was significantly inhibited when the concentration of acetic acid was higher than 1.5 g/L [66]. In addition, Zhang et al. indicated that AD was greatly inhibited by up to 17 g/L VFAs under a high OLR (8 g-VS/L-d) and failed within 10 days [67]. Furthermore, Gebreeyessus et al. found that MPA could tolerate 2.4 g/L acetic acid while 0.9 g/L propionic acid significantly inhibited MPA activity [68]. Therefore, acetate and propionic acid play important roles in biogas production, and their concentrations are defined as indicators of the performance of AD reactors [69].

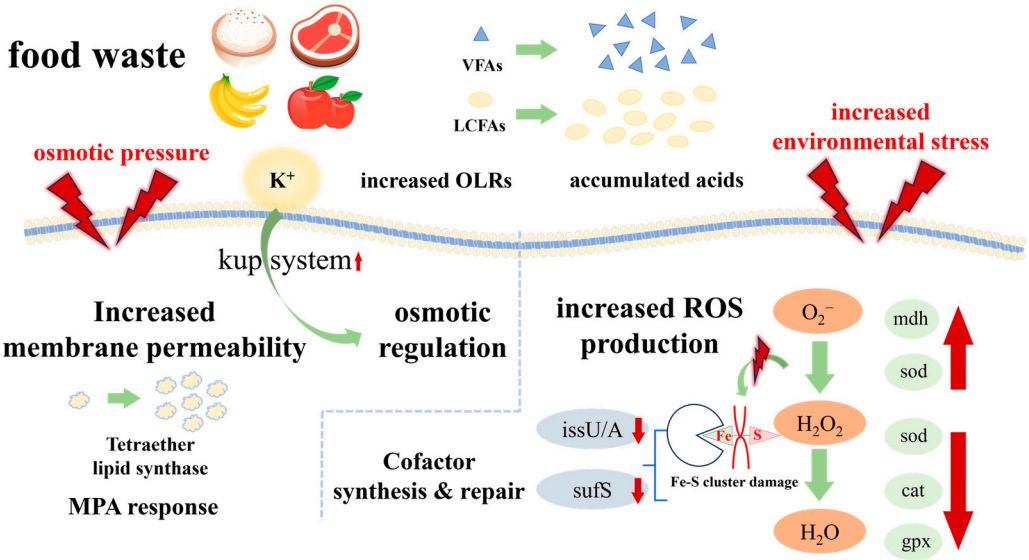


Figure 3. Increased OLRs result in VFA accumulation, consequently triggering microbial responses in terms of osmotic regulation, ROS production, and antioxidant defense metabolism. Abbreviations: proU, osmoprotectant transport system substrate-binding protein; mdh, malate dehydrogenase; cat, catalase; gpx, glutathione peroxidase; sod, superoxide dismutase; issU, nitrogen fixation protein; sufS, cysteine desulfurase.

During the acidogenic process, the degradation equations for propionic and butyric acids (shown below) demonstrate that their degradation processes exhibit high Gibbs free energies, thereby classifying them as thermodynamically non-spontaneous reactions.

The degradation reactions of propionic and butyric acids are only possible when the concentration of the degradation products is low and when the partial pressure of hydrogen is low [70]. However, the inhibition of MPA following the occurrence of acid inhibition results in an inability to rapidly utilize hydrogen, thereby perpetuating the production of hydrogen and acetic acid. Consequently, hydrogen partial pressure remains elevated, which impedes the decomposition processes of propionic and butyric acids and hinders the eventual manifestation of acid inhibition.

$$CH_3CH_2COOH + 2H_2O \rightarrow CH_3COOH + CO_2 + 3H_2 \quad \Delta G = +76.1\ kJ/mol \quad (1)$$

$$CH_3CH_2CH_2COOH + 2H_2O \rightarrow 2CH_3COOH + 2H_2 \quad \Delta G = +48.1\ kJ/mol \quad (2)$$

VFAs can be categorized into ionic VFAs and molecular VFAs (free VFAs). Ionic VFAs are unable to permeate the microbial membrane and therefore have minimal impact on the cell. Conversely, VFA accumulation will lead to a decrease in pH, which can promote the conversion of ionic VFAs into molecular ones. The lipophilic characteristics of molecular VFAs allow them to freely traverse the cell membrane and dissociate H^+ in the cytoplasm, contributing to cytoplasmic acidification. As a coping mechanism, the cell expels H^+ outward by active transport through the proton pump, which requires significant energy expenditure and diminishes microbial viability [71]. In addition, free VFAs penetrate the cell membrane and serve as coupling agents to impede microbial activity [72]. The intracellular pH gradually decreases with the accumulation of H^+, and the cell stops growing once the pH decreases to its lower limitation limit [73]. Consequently, free VFAs not only inhibit MPA but also suppress acid-producing bacteria. The inhibition concentration of acid-producing bacteria is higher than that for MPA. Specifically, the inhibition concentration of free VFAs for acid-producing bacteria was in the range from 2400 to 3000 mg/L [68].

Furthermore, VFAs can accumulate in AD processes operated at a high OLR, contributing to the presence of inhibition because functional microorganisms cannot rapidly degrade hydrogen, carbon dioxide, and VFAs. The metabolization of acetate produced by acetogenic bacteria relies on a sufficient increase in the number of MPA. Therefore, the accumulation of VFAs can lead to heightened activity of acid-producing bacteria and suppressed activity of MPA, which contributes to its concentration as an indicator of the reactor performance. Consequently, elevated levels of VFA accumulation hinder methanogenesis and induce microbial stress through rapid acidification, ultimately leading to deterioration of the AD process. Meanwhile, alterations in pH can impact the efficacy of acid inhibition by altering the morphology of VFAs. Cotter et al. observed that low toxicity from VFAs to cells was present at a neutral pH while more and more VFAs were converted from ionic to free VFAs with a decreasing pH; thus, the toxicity from VFAs increased drastically. In addition, AD bioreactor performance was impeded when the pH was at around 6.0 due to the accumulation of free VFAs, while adjusting the pH value to 8.0 restored reactor performance [74]. pH can also change the metabolic pathway of the acidogenic process, which in turn leads to a transformation in the types and concentrations of VFAs. For instance, Jiang et al. found that the main product of the acidogenic process was acetic acid when the pH was 5.0, while butyric acid became predominant when the pH was in the range of 6.0 to 7.0 [75].

4.1.2. Inhibition of LCFAs

LCFAs are defined as inhibitors in the AD system for FW treatment as well. During the FW hydrolysis process, lipids can be hydrolyzed rapidly to produce LCFAs and glycerol. In general, low concentrations of LCFAs can be degraded to acetic acid and release hydrogen through the β-oxidation process, while glycerol can be degraded into propylene glycol, which can then be converted to acetic acid. Subsequently, hydrogen and acetic acid can be

utilized by MPA for ultimate conversion to methane and carbon dioxide [76]. However, high concentrations of LCFAs can inhibit the methanogenesis process. With regard to different types of LCFAs, including saturated and unsaturated ones, unsaturated LCFAs are theoretically more obstructive to anaerobic microorganisms than saturated ones due to their faster accumulation rate on microbial cell membranes. Specifically, in previous research, researchers compared the inhibition of unsaturated oleate with saturated stearate and palmitate on anaerobic seed sludge, and they found that 1 mM oleate or more than 4 mM stearate and palmitate could inhibit 50% of methanogenic activity of *Methanobacterium formicicum* [77]. In terms of anaerobic granular sludge, LCFAs can adsorb onto microbial cytomembranes, inhibiting substance transportation and causing irreversible toxicity that destroys the structure of granular sludge. Therefore, the settling performance of granular sludge will decrease. Duan et al. discovered that LCFAs enhanced the bubble separation rate of anaerobic particles and reduced the probability of bubble–bubble aggregation, which had a significant inhibitory influence on methane production by hindering the conversion of VFAs to methane [78]. Furthermore, Maria Gaspari's research illustrated the inhibitory mechanism of LCFAs on MPA from a microcosmic perspective. The accumulation of LCFAs can lead to sludge floatation and thus deteriorate reactor performance [79]. It has been observed that there is a strong correlation between LCFAs' toxicity toward MPA and the inoculum-specific surface area; specifically, granular sludge is subjected to relatively lower LCFA toxicity due to its smaller specific surface area compared with suspended or flocculated sludge [65].

Acetate-utilizing MPA are more sensitive to the inhibition of LCFAs than hydrogen-utilizing MPA. Specifically, the metabolism of acetic acid by acetate-utilizing MPA is significantly inhibited when the linoleic acid concentration is 30 mg/L, while hydrogen-utilizing MPA shows minor inhibition. Furthermore, the biochemical process of methane synthesis utilizing hydrogen is stable, and the detrimental effects of LCFAs on acetate-utilizing MPA can be minimized through phase separation. The detrimental effect of LCFAs on acetate-utilizing MPA can be minimized through phase separation [80]. Therefore, a two-phase AD process is defined as more suitable for LCFA-rich substrates.

4.2. Solutions to Acid Inhibition

4.2.1. Alkaline Agent Addition

The addition of alkaline agents such as $Ca(OH)_2$, NaOH, and KOH to adjust the pH value in the AD system is the most typical method for alleviating acid suppression. This is beneficial for maintaining a stable and near-neutral pH environment to improve the metabolic activity of MPA. According to a previous study, Yang et al. elucidated the effect of pH adjustment through the addition of NaOH on the AD process. The results indicated that adding NaOH could effectively improve the utilization of acetic acids, alleviate LCFA accumulation, and increase methane production [81]. Furthermore, Zhang et al. improved the buffering capacity by adjusting the system's pH to around 7.5 with the addition of NaOH, which effectively prevented VFA accumulation and achieved a stable performance at a higher OLR [82].

Nevertheless, the addition of alkaline agents has limitations and can only operate when acid inhibition is not severe. It is not beneficial to add a large quantity of alkaline substances when the system experiences severe acidification because high concentrations of Ca^{2+}, Na^{+}, K^{+}, and Mg^{2+} can inhibit microbial activity as well. For instance, Wang et al. investigated the effects of metal cations on AD for FW treatment and found that low concentrations of Ca^{2+} and Mg^{2+} can promote the hydrolysis and acidification process, leading to increased cumulative methane production, while low concentrations of Mg^{2+} accelerated β-oxidation to alleviate acid accumulation [83]. In addition, when superfluous

NaOH was added to adjust pH, high concentrations of Na^+ (>4.0 g/L) inhibited MPA activity [84]. Meanwhile, Dai et al. showed similar results in that high concentrations of Na^+ (8.0 g/L) would severely impede the mesophilic AD process [85]. Another shortcoming of this approach is the requirement of substantial amounts of chemical substances, which results in heightened operating expenses.

4.2.2. Trace Element Addition

Trace elements such as Fe, Co, Mo, and Ni are crucial for MPA because the synthesis of certain key enzymes in microbial cells and the realization of methanogenesis functions depend on their presence. Therefore, insufficient concentrations of these trace elements in FW can limit microbial protein synthesis as well as normal physiological and metabolic functions [22]. All methanogenesis pathways ultimately lead to the reduction of methyl coenzyme M to methane by methyl coenzyme reductase and coenzyme F_{420}, which is a cofactor of methyl coenzyme reductase. Methyl coenzyme M containing Ni is a low-molecular-weight substance. Furthermore, Co plays an important role in methyl coenzyme M methyltransferase and cochleorin-like substances [86]. Wei et al. elucidated the effect of adding trace elements including Fe, Co, and Ni on AD systems for FW treatment; the results indicated that these trace elements could promote the synthesis of certain key enzymes in MPA to accelerate methane production and mitigate acid inhibition [87]. However, in actual the AD process, bioavailability and interactions between different elements can vary for trace elements. Therefore, the addition of trace metal elements involves two phases: a solid phase and a liquid phase, with complex interactions between them and other elements, contributing to the incomplete utilization. In addition, antagonistic effects between different elements can inhibit AD reactors if the trace elements are not properly matched [22].

4.2.3. Microbial Electrochemical and Biofortification Technology

In recent years, with the advancement of microbial electrochemical technology, researchers have proposed the introduction of electrochemical systems or conductive materials into AD systems to enhance system stability and methane production [88]. On the one hand, additional energy can be provided to microorganisms by introducing an electrochemical system to address the energy barrier problem caused by propionic and butyric acid accumulation. For instance, Hou et al. combined a traditional AD bioreactor with an algal microbial fuel cell phase (AMFC) to construct an AD-AMFC system, and this combined AD-AMFC system could accelerate AD efficiency, alleviate VFA inhibition, and increase methane production compared with the initial AD reactor [89]. In addition, Shi et al. developed a two-stage gas membrane absorption anaerobic reactor for the stable operation of FW treatment because it can mitigate VFA accumulation [90]. Park et al. integrated a microbial electrolysis cell (MEC) with an AD reactor to build an MEC-AD system that demonstrated enhanced the methane production rate and reduced the system stabilization duration. In comparison to traditional AD systems alone, the MEC-AD system can effectively enhance COD removal, accelerate VFA degradation, and improve system stability [91]. On the other hand, adding electrically conductive materials such as granular activated carbon, graphene, and magnetite into the reactor enables a relationship to be established between electroactive microorganisms and MPA through intermediate electron transfer mechanisms, thereby accelerating VFA degradation and enhancing the methane yield. Zhu et al. demonstrated that adding biochar—particularly that obtained from 600 °C pyrolysis—could enhance direct intermediate electron transfer (DIET) and alleviate acid accumulation in FW treatment under a high OLR [92]. Similarity, Yamada et al. discovered that introducing magnetite in a thermophilic AD reactor facilitated electron transfer processes between organic acid-degrading

bacteria and MPA while contributing to an accelerated conversion rate of acetic and propionic acid to methane while effectively mitigating the inhibitory effect on acid accumulation in the AD bioreactor [93]. Li et al. demonstrated that the adding biochar facilitated the construction of an electron transfer chain between organic acid-oxidizing bacteria and MPA, resulting in alleviated VFA accumulation, while a shortened lag phase for methane production increased methane production rates [94]. It has been observed that among various conductive materials, biochar is often selected as the preferred material due to its superior reusability and cost-effectiveness. Conversely, metal–organic framework (MOF) materials have also been employed as cathode electrodes, as illustrated in Figure 4. The distinctive physicochemical characteristics of the MOF cathode can facilitate the formation of a highly active cathode biofilm, predominantly composed of hydrogenotrophic MPA. The cathode's direct electron transfer and hydrogenotrophic methanogenesis pathways were reinforced, resulting in a reduction in the hydrogen partial pressure within the system. Furthermore, the microbial electrolysis cell coupled AD and metal–organic framework systems facilitated the enrichment of additional populations of MPA, thereby enhancing the carbon dioxide reduction pathway. The primarily enhanced enzymes included formylmethanofuran (EC:1.2.7.12); methenyltetrahydromethanopterin cyclohydrolase (EC:3.5.4.27); and 5,10-methylenetetrahydromethanopterin reductase (EC:1.5.92.8) [95]. In addition, biochar has a large specific surface area and distinctive physicochemical properties, which can buffer the issue of a pH drop caused by VFA accumulation and increase the alkalinity of the system. Nevertheless, the introduction of electrochemical systems and conductive materials into AD systems to enhance system stability and methane production efficiency is still in the early research stage, with most of the materials being tested using synthetic wastewater under a single mode of operation. It is recommended that researchers should further explore the actual treatment effects by regulating various parameters (e.g., cathode/anode potentials, electrode materials, etc.) at a later stage.

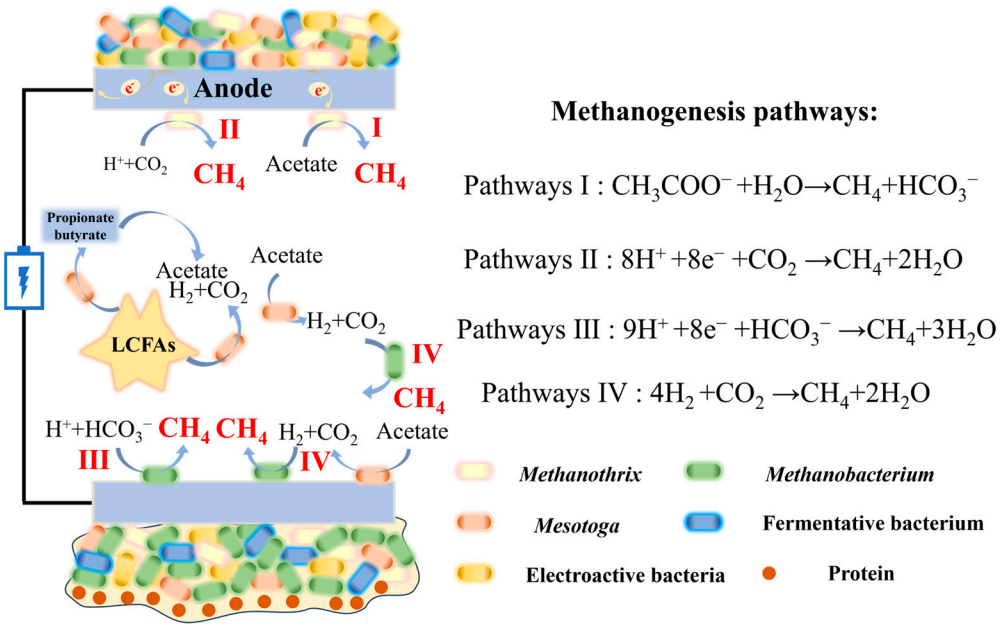


Figure 4. Mechanism diagram of MOF cathode promoting methane production from LCFA.

Biofortification involves the direct addition of a specific functional microorganism to AD systems to alleviate acid accumulation. The addition of a fortifying agent to an acid-suppressed digestion system can potentially shorten the recovery time and increase resistance to acid suppression. At present, numerous studies have been conducted on the

application of biofortification technology to relieve acid suppression and improve biogas production. For example, Li et al. improved the stability of the AD process in an overloaded reactor by introducing acid-tolerant methanogenic flora, which effectively relieved the accumulation of VFAs as well as acid inhibition [88]. Meanwhile, Zhao et al. innovatively proposed the application of "acid-eating and alkali-releasing" photosynthesizing bacteria to mitigate acid inhibition, observing significant relief within 12 days and a recovery to a normal gas production performance and effluent index in the reactor. Compared with other bio-enhanced microorganisms, photosynthetic bacteria are easy to enrich and cultivate, exhibiting rapid growth that allows them to reach logarithmic expansion within 5–10 days for expedited relief from acid suppression [96].

4.2.4. Micro-Aeration-Based AD Technology

In contrast to earlier research, which indicated that traditional AD required isolation from oxygen, recent studies have demonstrated that trace amounts of oxygen can facilitate the performance of AD bioreactors for FW treatment. The current mainstream method is to use the oxidation–reduction potential (ORP) to control the oxygen dosage. When the oxidation–reduction potential is in the range of −300 to 0 mV, it can be defined as a micro-aerobic environment, which is also the mainstream approach to controlling the oxygen intake at present. Specifically, this research has shown that oxygen can increase the activity of MPA, which in turn promotes acid utilization efficiency. It has been proven that different functional microorganisms can form microbial aggregates, which can be categorized as flocs, granules, and biofilms [97]. These flocs consist of MPA cores and bacterial surfaces, which enable them to withstand hostile environments and promote commensalism. Specifically, in micro-aeration systems, parthenogenetic anaerobes (e.g., Pseudomonas spp. in the phylum Firmicutes) with a high potential to consume oxygen were enriched, and their growth accelerates the hydrolysis and acidification phases, which in turn enriches acetate-utilizing MPA such as Methanosaeta to become the dominant MPA in the methanogenesis phase [98]. Furthermore, the addition of carriers and micro-aeration facilitated AD by inhibiting ammonia and nitrogen accumulation, which in turn enabled their reversion [99]. Wu et al. have demonstrated that the combination of carriers and micro-aeration can regulate carbon and nitrogen metabolism while mediating the stress responses. This can be ascribed to multiple responses to micro-aeration stress stimulation, which effectively promotes recovery from ammonia nitrogen inhibition [100]. The introduction of micro-aeration into FW AD systems promoted the electron transport system in the AD system, which in turn enhanced the key enzymes (e.g., amylase, α-glucosidase) associated with acidification. It can further enhance the performance of the bioreactor, particularly in reducing the accumulation of LCFAs in the treatment of lipid-rich organic waste [101]. Nevertheless, it has been demonstrated that the direct introduction of air into AD reactors results in accelerated acid accumulation. Consequently, Li et al. circumvented the shortcomings associated with direct air intake and promoted hydrolysis acidification and methanogenesis processes while enhancing performance through measured-flow micro-aeration. Additionally, they facilitated the conversion of short-chain fatty acids to acetic acid and established acetate-utilizing methanogenesis as the predominant methanogenesis pathway [98]. In conclusion, micro-aeration represents a novel approach to promoting traditional AD systems. However, when employing micro-aeration as a mitigation and promotion program, it is essential to select the most appropriate aeration schemes and dosage [99].

4.2.5. Overall Summary

The inhibition of acid accumulation (VFAs and LCFAs) has posed a critical challenge for AD under a high OLR. Various strategies have been developed to mitigate this issue, each with distinct mechanisms and performance characteristics. Table 3 summarizes the key solutions to acid inhibition, outlining their mechanisms, advantages, limitations, and supporting references, providing a comprehensive overview of the technological landscape and operational trade-offs.

Table 3. Solutions for acid inhibition in AD.

Strategy	Mechanism	Advantages	Limitations	Ref.
Alkaline Agent Addition	Adjusts system pH to neutral	Simple operation, low cost, and rapid relief of acid inhibition	Continuous addition incurs high costs	[81,82,84,85]
Trace Element Addition	Accelerate methanogenesis	Small additions enhance enzyme activity	Potential antagonistic effects between elements	[22,86,87]
Microbial Electrochemical Technology	Enhance direct interspecies electron transfer	Highly effective in relieving VFA accumulation	High initial investment	[88,89,91,92,95]
Biofortification Technology	Accelerate VFA degradation and pH recovery	Rapid system stability recovery	Lacking long-term stability	[88,96]
Micro-aeration	Optimizing microbial community structure	Reduce the accumulation of LCFAs, without the need for chemical additives	Excessive oxygen inhibits anaerobic microbial communities	[98,100,101]

5. Salt

5.1. Salinity Inhibition

Salt in FW mainly contains calcium, magnesium, sodium salts, etc. Sodium salt is defined as an indispensable flavoring agent in food processing, so it has the highest content in FW and the greatest impact on AD performance. Within a certain concentration range, salt can act as an inhibitor of the AD process for organic matter treatment. In the AD process for FW treatment, low concentrations of sodium salt (<5 g/L) can maintain the balance of microbial membranes and regulate osmotic pressure, which can promote enzyme activity and improve hydrolysis and acidification processes. Conversely, high concentrations of sodium salt (>5 g/L) can contribute to the dewatering of cultivated sludge and reduce its activity, which inhibits the AD process, as evidenced by decreased biogas production [102]. The inhibitory mechanism of a high salt concentration can be categorized into four main aspects: (a) reduction in sludge floc size and inhibition of filamentous bacteria; (b) cell rupture leading to a decrease in microbial activity; (c) alterations to enzyme structure and competition between binding sites; and (d) down-regulation of key genes, as presented in Figure 5 [103]. Specifically, high salinity impedes the diffusion of organic matter and disrupts bacterial growth, further affecting the physical properties and apparent morphology of sludge. In addition, high salinity greatly reduces enzyme activity because some metal ions can substitute for ions in metalloproteinase binding sites, rendering the enzyme structure undesirable afterwards [104].

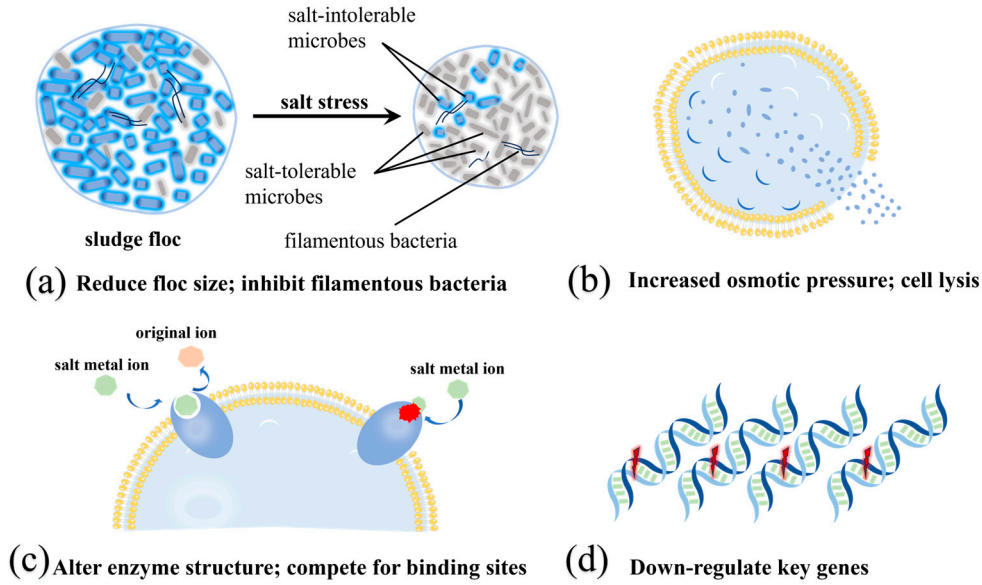


Figure 5. Mechanism of high-salinity inhibition in AD system.

Previous studies have demonstrated that the presence of Na^+ in FW influences the maximum amount of methane production [105]. During the operation process, the utilization of circulating water and increased feed concentration contribute to a cumulative increase in sodium salt concentration, impacting the normal function of AD systems. As a result, the impact of sodium salt on the AD performance has garnered attention from various researchers. Table 4 presents the impact of sodium salts on different stages of AD in recent research. Specifically, an excessive concentration of Na^+ can contribute to a rapid increase in the cell osmotic pressure and further result in cell plasma membrane separation, subsequent cell dehydration or rupture, and the ultimate reactor inhibition [106]. Furthermore, Yin et al. demonstrated that a low sodium salt concentration (0.6%) favored the methanogenesis process, contributing to a 12.2% increase in methane production compared to the control group. Conversely, when the sodium salt concentration was raised to 5.2%, the experimental group indicated a 72.3% decrease in methane production compared to the control group, with significant inhibition of MPA and down-regulation of genes related to fatty acid formation, including ackA, pta, and ACOX, and methane production, including mcrA and mtaA [107]. In addition, Zhang et al. discovered that when the concentration of NaCl in FW was 15 g/L, there was a significant reduction in both the diversity of archaea and the total abundance of MPA. Specifically, the abundance of Methanosarcina decreased significantly while the phylum Firmicutes was reduced to 7.73%, indicating a strong inhibitory effect of NaCl addition on this phylum [108].

Table 4. Effect of salinity on the four stages of the AD process.

Stage	Salinity	Results	Ref.
Hydrolysis and acidogenesis	Salinity: 0 g/L–8 g/L	VFAs: 367.6 mg-COD/g-VSS-638.5 mg-COD/g-VSS	[109]
	Salinity: 0.6%	Production of VFAs peaks	[107]
	NaCl: 0 g/L–5 g/L	VFAs: 15,624–20,316 mg/L	[106]
	NaCl: 5 g/L–15 g/L	VFAs: 20,316–5230 mg/L	
	NaCl: 2 g/L–15 g/L	Soluble proteins in the hydrolysis stage: 2156–3124 mg/L Soluble carbohydrates in the hydrolysis stage: 8596–12,054 mg/L	

Table 4. *Cont.*

Stage	Salinity	Results	Ref.
Acetogenesis	NaCl: 2.41 g/L–10.14g/L	Hydrogen production activity is inhibited: 30–90%	[110]
	NaCl: 0.0, 0.5, 1.0, 1.5, 2.0, 2.5% and 3.0%	Hydrogen production decreases with increasing concentration	[111]
	NaCl: >20 g/L	Acetogenesis stages are severely inhibited	[112]
	NaCl: >18 g/L	Acetogenesis stages are severely inhibited	[109]
Methanogenesis	NaCl: 0, 0.05, 0.2, 0.3 and 0.5 mol/L	Methane production decreases as the concentration increases	[113]
	NaCl: 20 g/L	Methanogenesis stage is completely suppressed	[114]
	NaCl: 6 g/L–22 g/L	Methane production decreases with increasing concentration	[115]
	NaCl: >16 g/L	Methanogenesis stage is completely suppressed	[83]
	NaCl: >8 g/L	When salinity further increased, the concentration of coenzyme F_{420} decreased dramatically; at the same time, the dehydrogenase concentration decreased significantly with increasing salinity	[116]

Similarly, magnesium salt is an essential component in FW, and its inhibition mechanism for AD is approximately the same as that of sodium salt. Nevertheless, the effect of magnesium salt on anaerobic sludge differs slightly in terms of promotion and inhibition. Specifically, previous studies elucidated the effect of magnesium chloride on different phases of AD, demonstrating that low concentrations of magnesium chloride (<50 g/L) significantly promoted the hydrolysis during the acid-producing phase, resulting in a 265.2% increase in soluble SCOD compared to that of the control group when the concentration was 50 g/L. Conversely, high concentrations of magnesium chloride (>50 g/L) resulted in serious acid accumulation, and MPA were more sensitive to magnesium chloride, leading to a reduction in their activity [115]. Similar conclusions were obtained by Zhao et al., which indicated that methane production was only slightly lower than that of the control group when the concentration of magnesium chloride was less than 20 mg/L, and then decreased to only 50% of initial methane production when increased to 30 mg/L, and finally ceased completely at a concentration of 50 mg/L [117].

Sulfur is an essential element required by MPA, and their cells contain more sulfur than other anaerobic microorganisms. Sulfate-reducing bacteria (SRB) are responsible for reducing sulfate to sulfide and they can play crucial roles in the AD process for complex substrate biodegradation. When sulfate is introduced into an anaerobic bioreactor, SRB can convert sulfate into sulfide (S^{2-}), resulting in severe inhibition of MPA activity, which reduces methane production by competing for available hydrogen and carbon sources. Therefore, the reduction of sulfate by SRB is a key factor influencing the performance of AD. In addition, competition for substrates between SRB and MPA can constitute an inhibition because sulfate can be reduced by SRB. Therefore, keeping S^{2-} concentration below the assessable level is essential to maintain the stability of the anaerobic bioreactor. Sulfide toxicity appears to be corelated with free hydrogen sulfide concentrations in the pH range of 6.4–7.2. Inhibitory sulfide levels were in the range of 100–800 mg/L dissolved sulfide or approximately 50–400 mg/L undissolved H_2S [118]. It has been observed that the toxicity of sulfide increases in conjunction with pH increases. H_2S is the primary toxic form of sulfide because it can permeate cell membranes, create disulfide cross-links between polypeptide chains, and denature proteins, thereby affecting cellular metabolism.

5.2. Solutions to Salinity Inhibition

Due to the limitations of desalination technology, effectively removing salt from FW is not feasible, resulting in the inhibition of the AD process due to the presence of salt. The most typical method of desalination at the laboratory scale is water washing. However, due to the huge amount of FW generated daily, water washing will undoubtedly increase the difficulty of AD. Therefore, it is necessary to reduce and eliminate the inhibition caused by salt on AD for FW treatment.

In addition to the water-washing method, adding conductive materials is also one of the conventional physical methods, which features a simple operation and no chemical residue. Tanisho et al. discovered that the conversion of NADH to hydrogen can take place at the cell membrane, implying that the hydrogenase involved in the NADH pathway is a membrane-bound enzyme. Due to the negative charge of NADH, air nanobubbles can facilitate the exchange of electrons between NADH ferredoxin oxidoreductase and hydrogenase to produce hydrogen, thereby relatively increasing the activity of the electron transport system (ETS) and coenzyme F_{420} through enhanced electron transfer stimulation between bacteria and MPA [119]. Furthermore, incorporating conductive materials can alleviate the inhibitory effect of salinity on AD for FW treatment. For example, at 1.5% salinity, powdered activated carbon and magnetite were beneficial in sustaining a high methane production efficiency by facilitating DIET and homoepitrophic oxidation of butyrate and propionate. Additionally, adding these substances provided microorganisms with additional energy to counteract salt inhibition [120].

Bio-electrochemical AD is a novel biological approach for enhancing AD performance. Previous studies have demonstrated the potential of microbial electrochemical technology in reducing acid accumulation and mitigating the inhibitory effects of salinity on the AD process. For instance, in previous experiments investigating the impact of Fe-C microbial electrochemical technology on AD performance for FW biodegradation, Qu et al. discovered that this technology contributed to an enhancement in salt-tolerant gene abundance and enzyme activities within microbial communities, while significantly improving the salinity tolerance performance of sludge at high salinity levels [105]. This method has strong sustainability and can adapt to high-salt environments for extended periods; however, it has high technical complexity and elevated electrode maintenance costs. Ammonium acetate, as an exogenous restorative factor, has been shown to effectively mitigate the inhibitory effect on AD for FW treatment under high-salinity conditions. Specifically, Liu et al. compared the process of AD under high salinity with or without restorative additives, demonstrating that ammonium acetate can effectively alleviate the negative impact of a high-salt environment on anaerobic microbial flora including MPA, contributing to a significant increase in cumulative methane production by about 30% [121].

In addition to electrochemical techniques, the addition of osmoprotectants is another method for enhancing the salinity tolerance performance of sludge. Specifically, Hou et al. discovered that in an air nanobubble water reactor with the addition of 0–30 g/L NaCl, the hydrogen yield increased by 21–65% compared with the corresponding deionized water group, followed by a 14–43% increase in methane yield. This study demonstrated that the addition of air nanobubble water enhanced the enzymatic activity of individual stages when both stages of FW AD were exposed to the same salinity level for the first time. Figure 6 shows the results of ETS activity, which further indicated that the air nanobubbles were involved in electron transfer associated with hydrogen synthesis. Air nanobubbles allowed the reaction of $NADH + H^+ \rightarrow DAH^+ + H_2$ to proceed more efficiently and enhanced the activity of extracellular hydrolases. They also enhanced coenzyme F_{420}, facilitated electron transfer associated with hydrogen and methane synthesis, and alleviated the inhibitory effect of salinity [112]. Similarly, the inclusion of osmoprotectants also aided in reducing

salinity's effect on AD performance. For instance, it was demonstrated that when the Na^+ concentration increased from 0 to 20 g/L, cumulative methane production decreased from 623.97 mL/g-VS to 0. However, when 2.0 and 2.5 g/L glycine betaine were added to the reactor with Na^+ concentrations of 5 and 10 g/L, respectively, methane production could be enhanced by 29.07% and 63.49%, respectively [122]. These chemical solutions can quickly relieve osmotic pressure and are highly targeted. However, their addition will contribute to high reagent costs, which may introduce secondary pollution and require cautious operation.

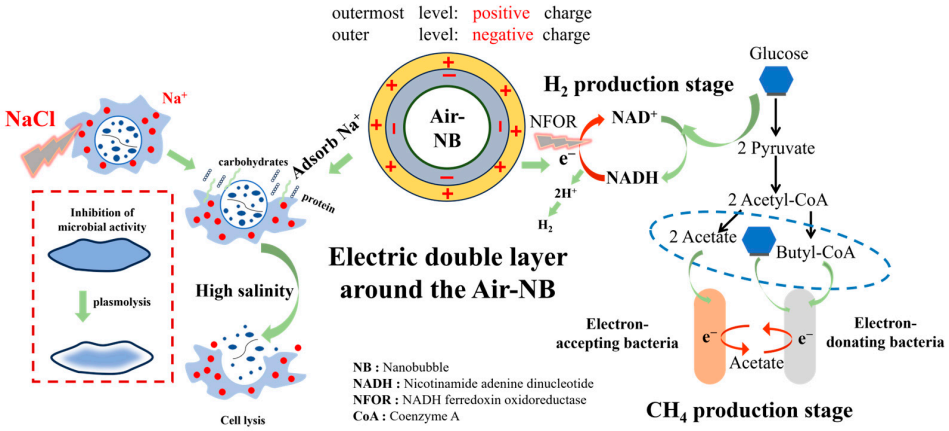


Figure 6. Impact of NaCl on microorganisms and mechanisms involved in the enhanced H_2 and CH_4 production from two-stage AD for high-salinity FW treatment with Air-NB addition and resultant promoted electron transfer.

6. Conclusions

During FW AD under a high OLR (>3 g-VS/L d), the AD process can be hindered by acid, ammonia nitrogen, and high-salinity inhibition caused by the inherent characteristics of FW. Among numerous mitigation strategies, co-digestion can achieve more balanced nutrient levels, alleviate ammonia nitrogen inhibition, and dilute toxic pollutants, while the application of microbial electrochemical technology can improve DIET, mitigate acid accumulation, and alleviate high-salinity inhibition to a certain extent. The introduction of additives and exogenous repair factors also shows enormous potential in enhancing microbial diversity, accelerating the synthesis of key enzymes for AD. Meanwhile, an ecological toxicity database for additives (such as nano-iron and osmoprotectants) should be established to prevent secondary pollution. High-salinity environments can impact AD microbial communities. The material costs, energy consumption, and maintenance expenses of laboratory strategies increase significantly after scaling up; however, the revenue from methane cannot cover the cost gap. In the future, it is necessary to combine Life Cycle Cost Analysis (LCCA) with a priority to develop low-cost technologies for "treating waste with waste" (such as co-digestion and waste-based adsorbent materials), rather than relying on high-input chemical or electrochemical methods.

7. Research Gaps and Opportunities

With regard to an FW AD system under a high OLR, research gaps still exist and opportunities are emerging simultaneously.

7.1. Research Gaps

1. Practical application: It is difficult to translate laboratory findings into practical applications. In actual plants, it is challenging to increase the OLR. Research on substrate selection for co-digestion and microbial communities is insufficient and new technologies are costly.

2. Environmental impact research: The environmental risk assessment of additives and exogenous remediation factors is incomplete, and the environmental impacts of combined mitigation strategies remain unclear.
3. Technical integration challenges: Existing mitigation strategies mostly target single inhibitory factors. The synergy and combination of different technologies are undefined, and their compatibility with existing processes needs to be explored.
4. Model prediction defects: Computer models struggle to accurately describe the dynamics of AD systems. Machine learning models are prone to overfitting, and the interpretability of these models is poor.

7.2. Research Opportunities

1. Technological integration and innovation: The integration of multiple mitigation strategies is expected to generate synergies, enhancing the efficiency and stability of AD.
2. Monitoring technology upgrades: Advanced monitoring technologies should be developed to accurately monitor key parameters and microbial community changes for refined control in real-time.
3. Sustainable development direction: Waste should be used as additives or co-substrates to explore environmentally friendly and economically viable mitigation strategies.
4. Expansion of application fields: The cross-integration of AD with other fields like biorefining should be promoted to expand its applications in multiple areas.

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Abbreviations

AD	anaerobic digestion
FW	food waste
OLR	organic loading rate
FAN	free ammonia
VFAs	volatile fatty acids
LCFAs	long-chain fatty acids
MPA	methane-producing archaea
CSTR	continuous-stirred tank reactor
SDSAR	siphon-driven self-agitated reactor
COD	chemical oxygen demand
ROS	reactive oxygen species
AMFC	algal microbial fuel cell phase
MEC	microbial electrolysis cell
DIET	direct intermediate electron transfer
MOF	metal–organic framework
SRB	sulfate-reducing bacteria
ETS	electron transport system

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